

The Security phase ensures your chatbot application implements comprehensive security controls at all layers. This section focuses on creating detailed security specifications that protect user data, prevent attacks, and maintain compliance with regulatory requirements.

What You'll Learn

In this security phase, you'll:

- **Implement authentication and authorization** with JWT tokens, RBAC, and session management
- **Design data protection** strategies with encryption, privacy controls, and GDPR compliance
- **Create application security** controls preventing common vulnerabilities and attacks
- **Establish compliance governance** with audit trails, incident response, and regulatory frameworks

Security Exercises

1. Authentication and Authorization

Create comprehensive authentication systems with JWT tokens, multi-factor authentication, role-based access control, and secure session management.

2. Data Protection and Privacy

Implement data protection strategies including encryption, privacy by design, GDPR compliance, and data lifecycle management.

3. Application Security Controls

Design security controls preventing OWASP Top 10 vulnerabilities, input validation, output encoding, and secure coding practices.

4. Compliance and Governance

Establish compliance frameworks for SOC 2, audit logging, incident response, and regulatory requirements.

Key Benefits

By completing the Security phase, you'll have:

- **Robust Authentication:** Multi-layered authentication with proper session management
- **Data Protection:** Comprehensive encryption and privacy controls
- **Vulnerability Prevention:** Protection against common web application attacks
- **Compliance Framework:** Audit trails and regulatory compliance procedures

Advanced Features Integration

Throughout this phase, you'll leverage:

- **Web Search MCP Server:** For current security best practices and threat intelligence
- **AWS Documentation MCP Server:** For AWS security service configurations
- **Agent Hooks:** For security validation and compliance checking
- **Agent Steering:** For secure coding standards and security requirements

Prerequisites

Before starting the Security exercises:

- All previous phases completed with security considerations
- Understanding of common web application vulnerabilities
- Familiarity with encryption and authentication concepts
- Knowledge of regulatory compliance requirements

Let's begin with comprehensive authentication and authorization systems that form the foundation of your application's security posture.